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Developmental changes in infants' sensitivity to vowel harmony for word segmentation

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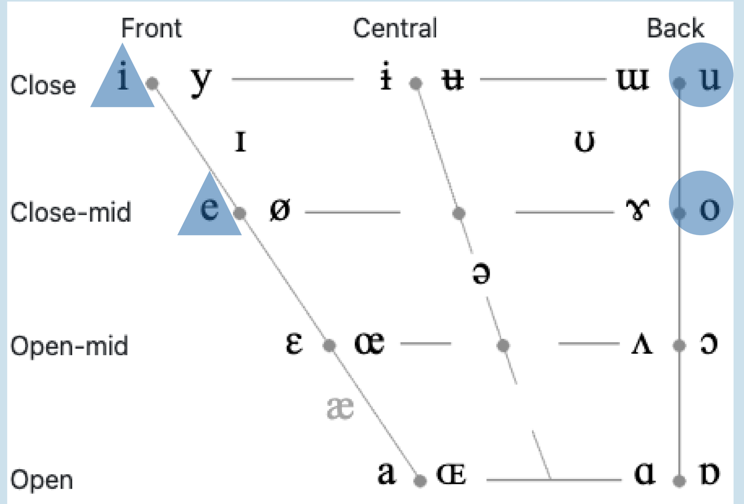
Introduction

- **Vowel harmony (VH)** is a phonological process in which vowels within a word agree on certain phonetic feature[1]
  - e.g., backness in Turkish *çocuk-lar* "children" *ev-ler* "houses"



- Disharmonies can signal word boundaries, making VH a potential cue for word segmentation[2]
- Infants learning VH languages (e.g., Turkish[3] and Akan[4]) distinguish harmonic vs. disharmonic words by 6 months, and use VH for word segmentation by 9-11 months[5]
- Recent evidence suggests an **early, broad sensitivity to VH**:
  - 4-month-old infants primarily exposed to non-VH languages prefer harmonic compared to disharmonic sequences[6]
  - 7-month-old English-learning infants (a non-VH language) can segment speech using VH cues[7]
- This sensitivity **attenuates without relevant input**:
  - By 7-8 months, infants exposed primarily to non-VH language(s) no longer prefer harmonic words[6,7], a pattern consistent with perceptual attunement
- Bilingual infants may follow a different developmental path[8]:
  - Experience with multiple languages could maintain broader perceptual sensitivity[9]
  - However, exposure to more than one non-VH languages may also accelerate the down-weighting of VH as a cue

Does VH sensitivity decline in infants with no exposure to VH, and does exposure to more than one non-VH language change this trajectory?

Participants

- 35 mono- and 36 bilingual 6-month-old infants
- 27 mono- and 19 bilingual 10-month-old infants
- All infants had no to minimum (< 5%) exposure to VH

Methods – Central Fixation

Baseline Phase (testing preference)

- 4 fixed-length trials with a disyllabic word repeated 16 times, with 500-ms interstimulus interval (ISI)

ke-mi, no-gu vs. ki-mo, nu-ge

Familiarization Phase

- 108-s continuous speech stream (45 repetitions) with no acoustic, prosodic, or statistical cues to word boundaries

de-bi-do-tu-be-pi-to-pu

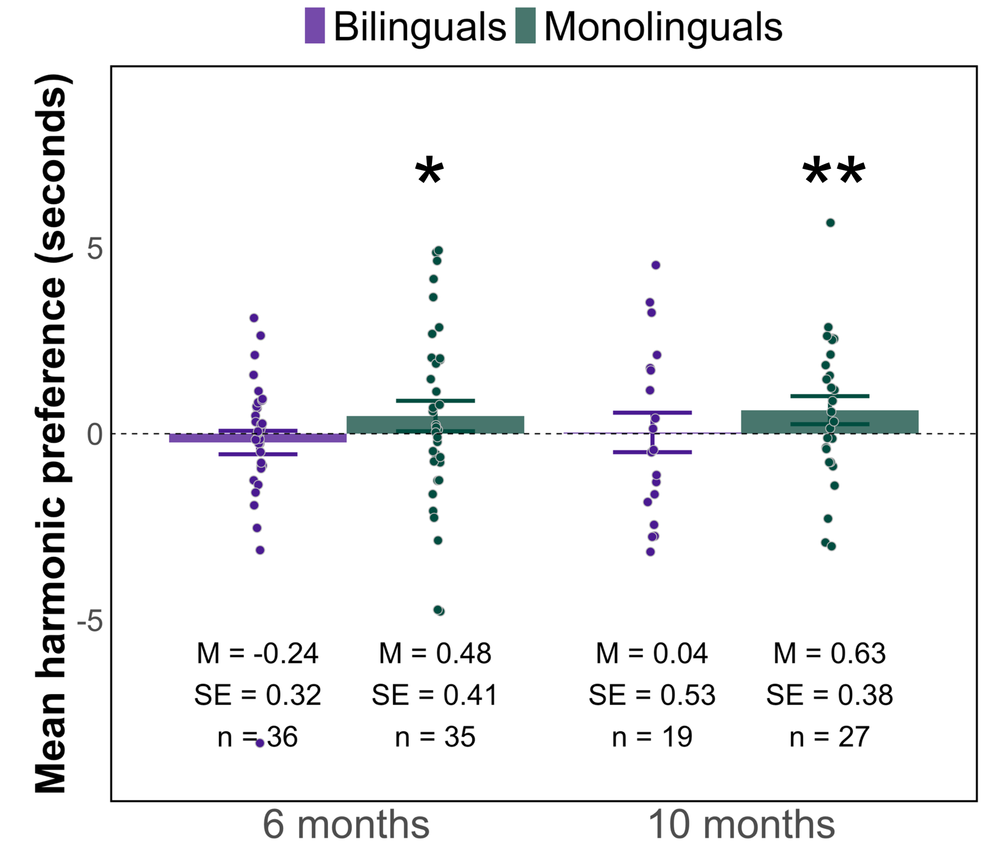
Test Phase (testing segmentation)

- 8 infant-controlled trials presenting a disyllabic sequence from the familiarization stream with 500-ms ISI

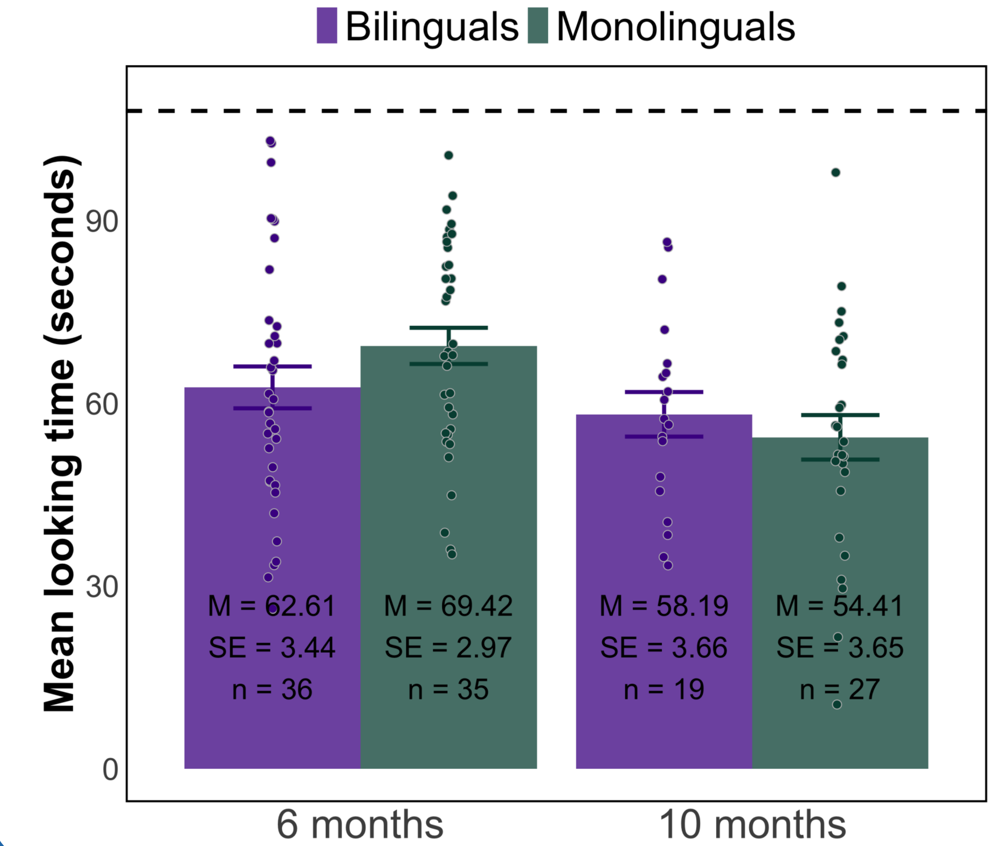
to-pu, de-bi vs. bi-do, tu-be

Preliminary Results

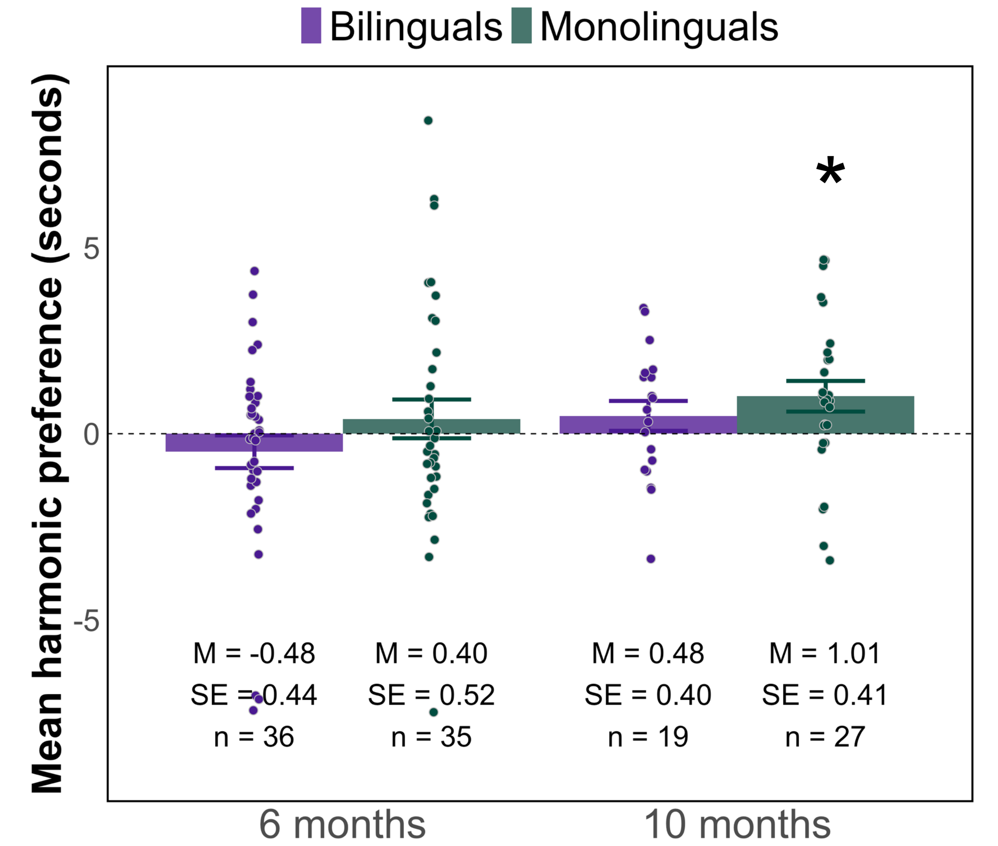
At baseline, infants in general showed a ***harmonic*** preference (p = .046), and this preference was more pronounced in the **monolingual** infants (p = .018).



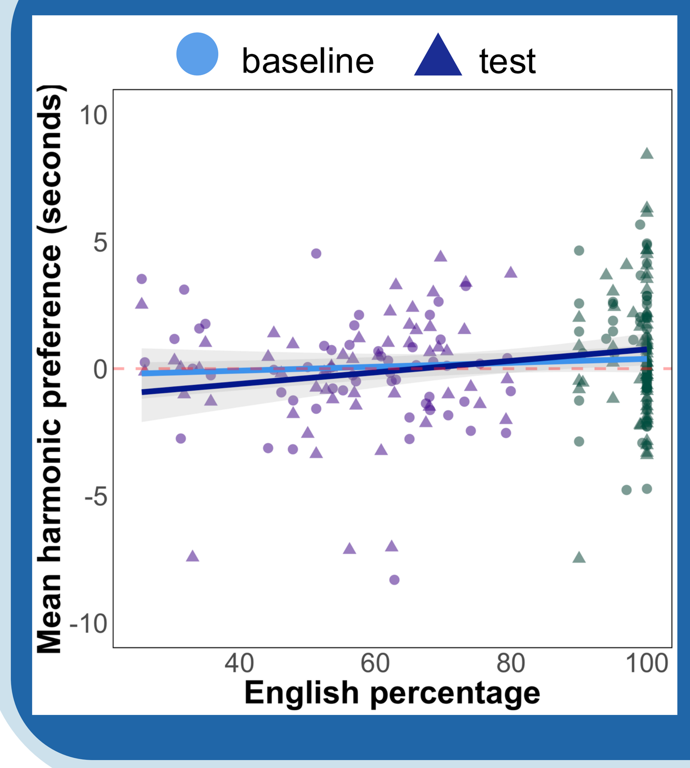
In familiarization, 10-month-olds looked shorter than the 6-month-olds (p = .007), and no significant interaction was observed for age x bilingualism.



At test, we did not find evidence for successful segmentation overall, but 10-month-old **monolinguals** showed a harmonic preference (p = .043).



Summary & Discussion



- English monolinguals preferred harmonic words at baseline:
  - Adding English exposure yielded an interaction with harmony at baseline (p = .093) and at test (p = .047)
  - Preliminary analysis of infant-directed corpus[10]: ~70% English disyllabic words agree in rounding
- No clear evidence for infants using VH for word segmentation: Persistent preference across phases in 10-month-old monolinguals obscures segmentation effects
- **Bilingual exposure**: Lack of VH sensitivity in bilingual infants suggest either (1) English input drives the sensitivity, or (2) bilingual exposure accelerates the decline in their VH sensitivity

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